Wang Yukun

Target Position: Assistant Researcher (Bioinformatics), Henan Academy of Sciences

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Education

ShanghaiTech University, School of Life Science and Technology 2022.09 – 2027.07 (Expected)

Ph.D. Candidate in Biology (Bioinformatics Track) | Supervisor: Prof. Liye Zhang

Henan University of Technology, School of Bioengineering 2018.09 – 2022.07

B.Eng. in Bioengineering (GPA 3.8/5.0) | Supervisor: Assoc. Prof. Jia Zhou

Research Interests

My research focuses on **bioinformatics** and **cancer genomics**, particularly the molecular subtyping and mechanistic dissection of hepatocellular carcinoma (HCC) recurrence and metastasis. I integrate genome, transcriptome, and DNA methylation data with **machine learning** approaches for multi-omics analysis, aiming to build a translation-oriented research programme on tumour evolution and therapeutic targets.

Publications & Academic Presentations

Peer-reviewed Publications

1. Zhou, X., Chen, C., Zeng, Y., Yang, Z., Zhuo, Y., **Wang, Y.**, Zhang, L., Cai, X., and Guo, X. (2025). Neoadjuvant Chemoimmunotherapy Combined With Node-Sparing Radiotherapy for Clinical T3N+ Locally Advanced Esophageal Squamous Cell Carcinoma: A Prospective Single-Arm, Phase II Study (CINSREC Trial). *Thoracic Cancer* 16(22), e70191. doi: 10.1111/1759-7714.70191
2. Song, M., Ma, S., Wang, G., **Wang, Y.**, Yang, Z., Xie, B., Guo, T., Huang, X., and Zhang, L. (2025). Benchmarking copy number aberrations inference tools using single-cell multi-omics datasets. *Briefings in Bioinformatics* 26, bbaf076. doi: 10.1093/bib/bbaf076
3. Zhou, J., Chen, Y., Qu, J.-H., **Wang, Y.-K.**, Mai, W.-N., Wan, D.-J., and Lu, X.-Y. (2022). Responses of microbial community and antibiotic resistance genes to co-existence of chloramphenicol and salinity. *Applied Microbiology and Biotechnology* 106, 7683–7697. doi: 10.1007/s00253-022-12188-3
4. Zhou, J., Chen, Y., Qu, J., **Wang, Y.**, Li, P., Xu, W., Huang, Y., and Mai, W. (2022). Isolation of a halotolerant *Halomonas* sp. A20 and its saccharin sodium wastewater degradation characteristics. *Microbiology China* 49, 942–955. doi: 10.13344/j.microbiol.china.210521

Manuscripts Under Review

5. **Wang, Y.**, et al. (2026). Cholangiocarcinoma-like phenotype defines a high-risk subclass of multicentric occurrence hepatocellular carcinoma. Submitted to *GUT*.

Conference Presentations

- 2025 — ShanghaiTech Life Science and Technology Frontier Conference, **Best Oral Presentation Award**; Annual Meetings of Shanghai Genetics Society and Shanghai Bioinformatics Society (Poster)
- 2024 — Annual Meeting of Shanghai Genetics Society (Poster)

Research Experience

1. Clonal Evolution and Multi-omics Mechanisms of HCC Recurrence and Metastasis

Principal Investigator (First Author) May 2023 – Present

- **Platform Construction:** Built a standardised multi-omics pipeline on a Linux HPC cluster (SLURM), integrating BWA / GATK / STAR for large-scale WES, RNA-seq, and RRBS processing, improving analysis throughput by ~40%.
- **Clonal Evolution Reconstruction:** Integrated somatic mutations and copy-number alterations (SCNA) and used PTI and Treeomics to reconstruct phylogenies of multicentric recurrent tumours; **first systematically defined** a highly aggressive clonal evolution pattern termed "**Reseeding**".
- **Cross-omics Phenotyping:** Applied ssGSEA and immune deconvolution to transcriptomic data, revealing that Reseeding HCC exhibits a marked "*cholangiocarcinoma-like (CCA-like)*" malignant reprogramming signature strongly associated with poor prognosis.
- **Mechanistic Validation:** Integrated genome-wide DNA methylation with transcriptional network analysis to identify **ESRP1** as a key splicing factor regulated by promoter hypomethylation, promoting **CD44v8** pro-metastatic alternative splicing; functionally validated in orthotopic xenograft models.
- First-author manuscript under review at *GUT*.

2. Medical Multimodal Foundation Models and Computational Pathology

Computational Method Development 2025 – Present

- **Single-cell / Spatial Foundation Models:** Applied the pretrained single-cell foundation model **Nicheformer** to single-cell and spatial transcriptomic data to build cell-level embeddings for cell-state representation and clustering, with cross-platform gene mapping and input-alignment workflows.
- **Pathology Foundation Models and MIL:** Extracted patch-level embeddings from H&E whole-slide images with the pathology vision foundation model **UNI2-H** and built weakly supervised multiple-instance learning models with **CLAM** for TCGA HCC / CCA-like / CHOL phenotype classification with cross-validated generalisation assessment.
- **Cross-modal Modelling:** Explored joint modelling of pathology image embeddings with transcriptomic / multi-omics features for tissue phenotype identification and biological subtype prediction.

3. Benchmarking CNA Inference Tools Using Single-cell Multi-omics Datasets

Contributing Author Jan 2024 – Dec 2024

- Systematically benchmarked 10+ single-cell CNA inference tools, quantifying accuracy, computational efficiency, and applicable scenarios; published in *Briefings in Bioinformatics* (IF = 9.5, Q1).

Technical Skills

Omics Analysis Standard analysis workflows for genome, transcriptome, and DNA methylation sequencing data; single-cell and spatial transcriptomics; phylogenetics and clonal evolution analysis

Programming & Tools Python, R, Bash / Linux HPC (SLURM); Git, Snakemake, LaTeX

AI / Machine Learning Machine learning for multi-omics data integration, feature selection, and survival risk modelling; representation learning and transfer of single-cell, spatial, and pathology image foundation models

Research Domains Cancer genomics (HCC / CCA), intratumour heterogeneity, bioinformatics method development

English CET-6 (507), fluent reading and academic writing

Awards & Honors

- 2025 — Outstanding Student (San Hao Student), ShanghaiTech University
- 2024 — Outstanding Communist Youth League Member, ShanghaiTech University